Microsoft

Microsoft Fabric Roadshow

Unify your data estate with OneLake, the OneDrive for data

Just Blindbæk

Principal Architect & MVP
twoday

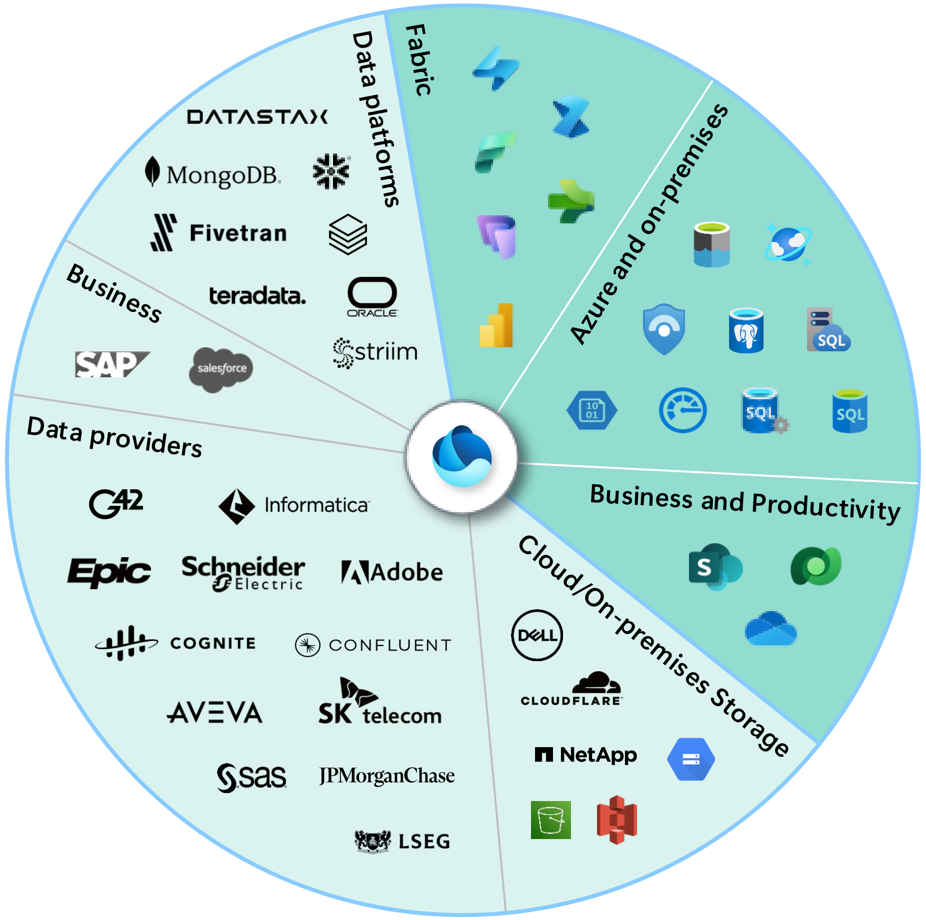
"OneLake spans all databases and clouds, including semantic models from Power BI. And therefore, it is the best source of knowledge and grounding for AI applications and context engineering."

Satya Nadella

A vast fragmented data landscape

Data comes from all over

OneLake unifies a complicated data landscape



All Fabric data is available in OneLake

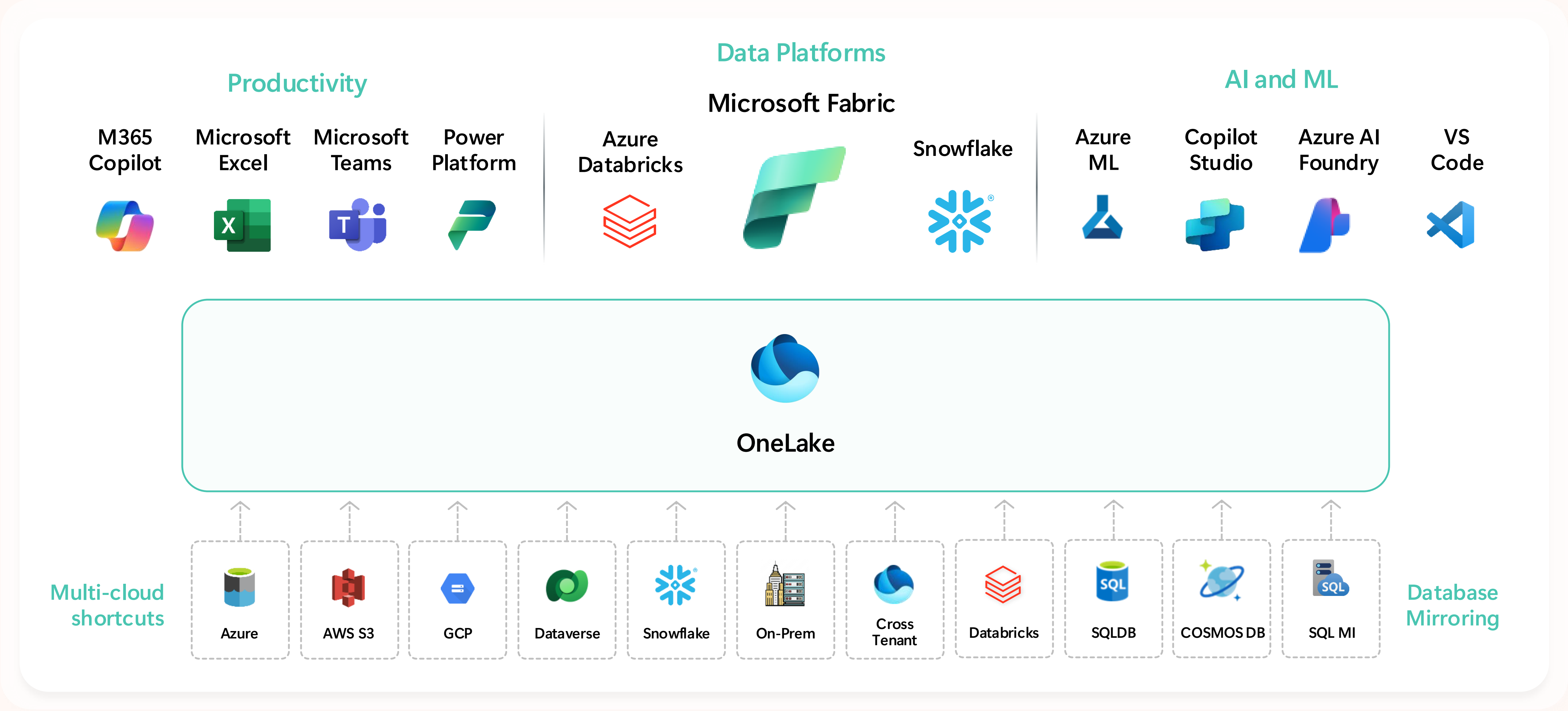
Microsoft data including Azure Data and Dynamics 365

Non-Microsoft data platforms and services

Data stored on-premises or cross cloud. In open or proprietary storage

Structured and unstructured data

OneLake data is available everywhere



Built on giants:

OneLake is storage at the core, but much more

ADLS

- Files & folders
- High scale
- Account & containers
- Hierarchical namespace
- Block blobs
- BCDR & high availability

Price: $0.023 per GB

Microsoft OneLake

- Multi-cloud shortcuts & mirroring
- Data sharing
- Unified namespace
- Catalog & Security
- Workspaces & Items
- Analytical tables

Price: $0.023 per GB

Microsoft OneLake

Open and AI-ready data lake

Unified data foundation for the era of AI

Unified, open access across platforms

Unified discovery and governance

Microsoft OneLake

Open and AI-ready data lake

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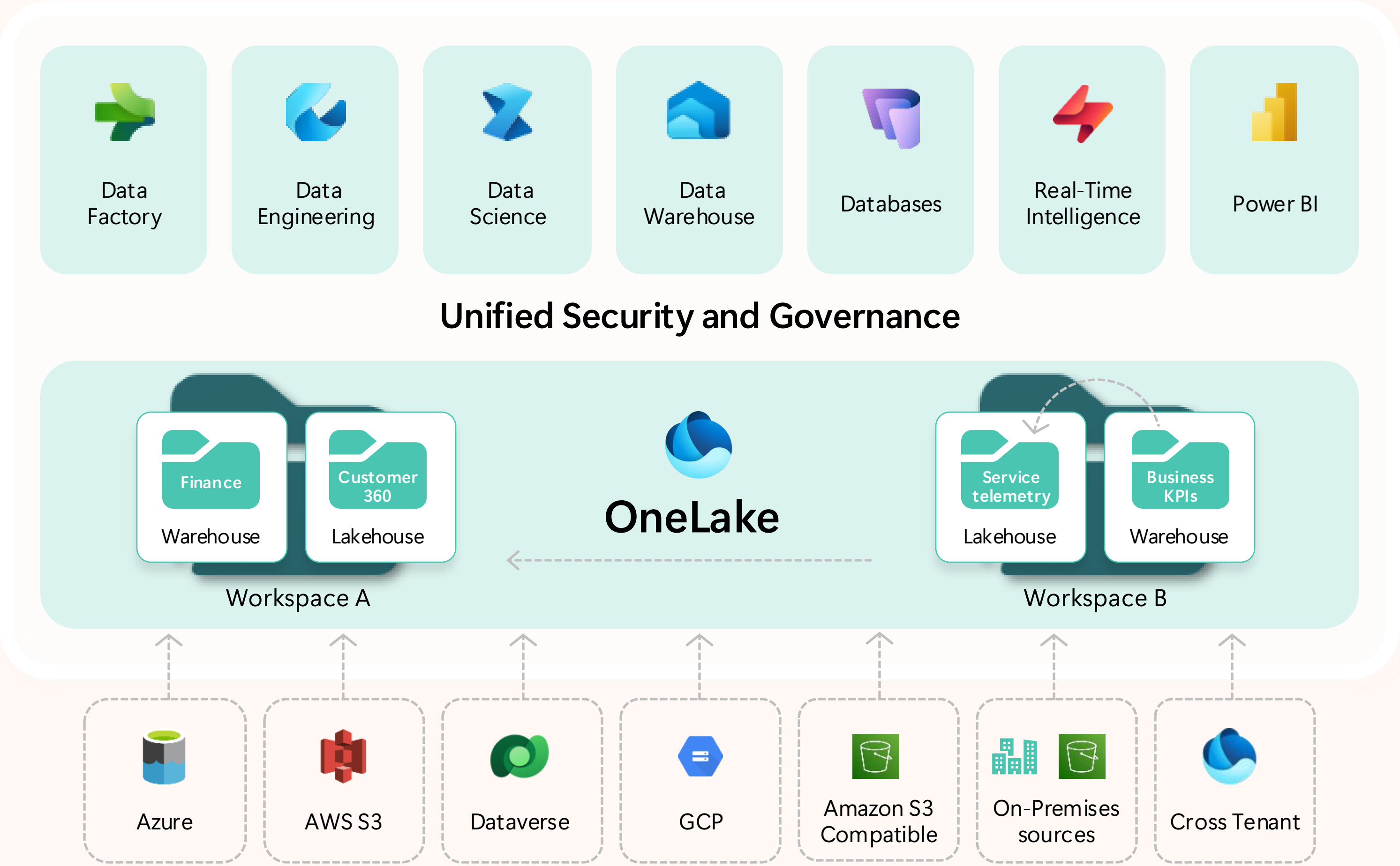
Unified discovery and governance

Shortcuts virtualize data across domains and clouds

No data movement or duplication



A shortcut is a symbolic link which points from one data location to another

Create a shortcut to make data from a warehouse part of your lakehouse

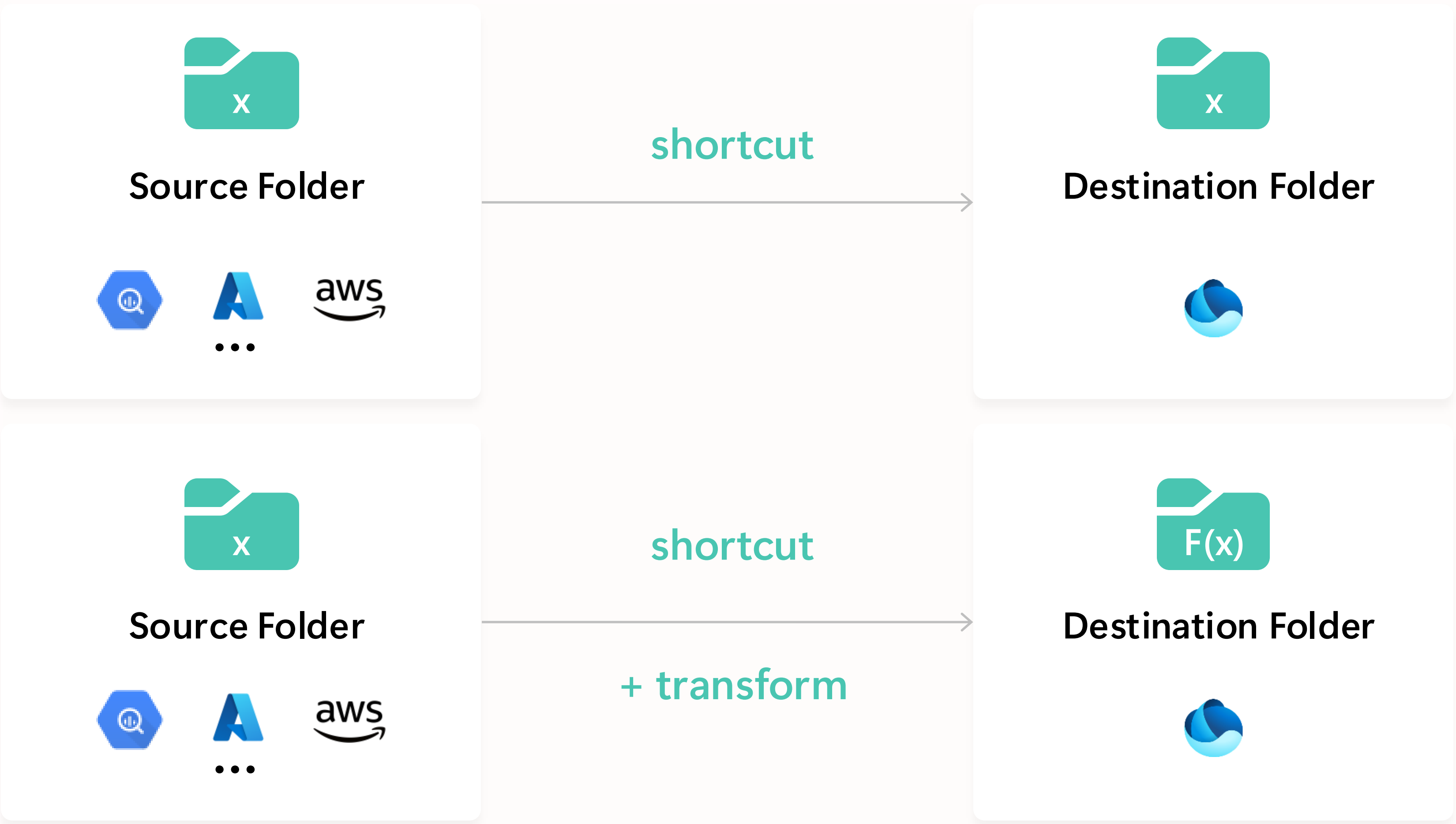
Create a shortcut within Fabric to consolidate data across items or workspaces without changing the ownership of the data. Data can be reused multiple times without data duplication

Existing ADLS Gen2 storage accounts and Amazon S3 buckets can be managed externally to Fabric and Microsoft while still being virtualized into OneLake with shortcuts

All data is mapped to a unified namespace and can be accessed using the same APIs including the ADLS Gen2 DFS APIs

Shortcut transformations

Seamless enrichment with native AI in OneLake

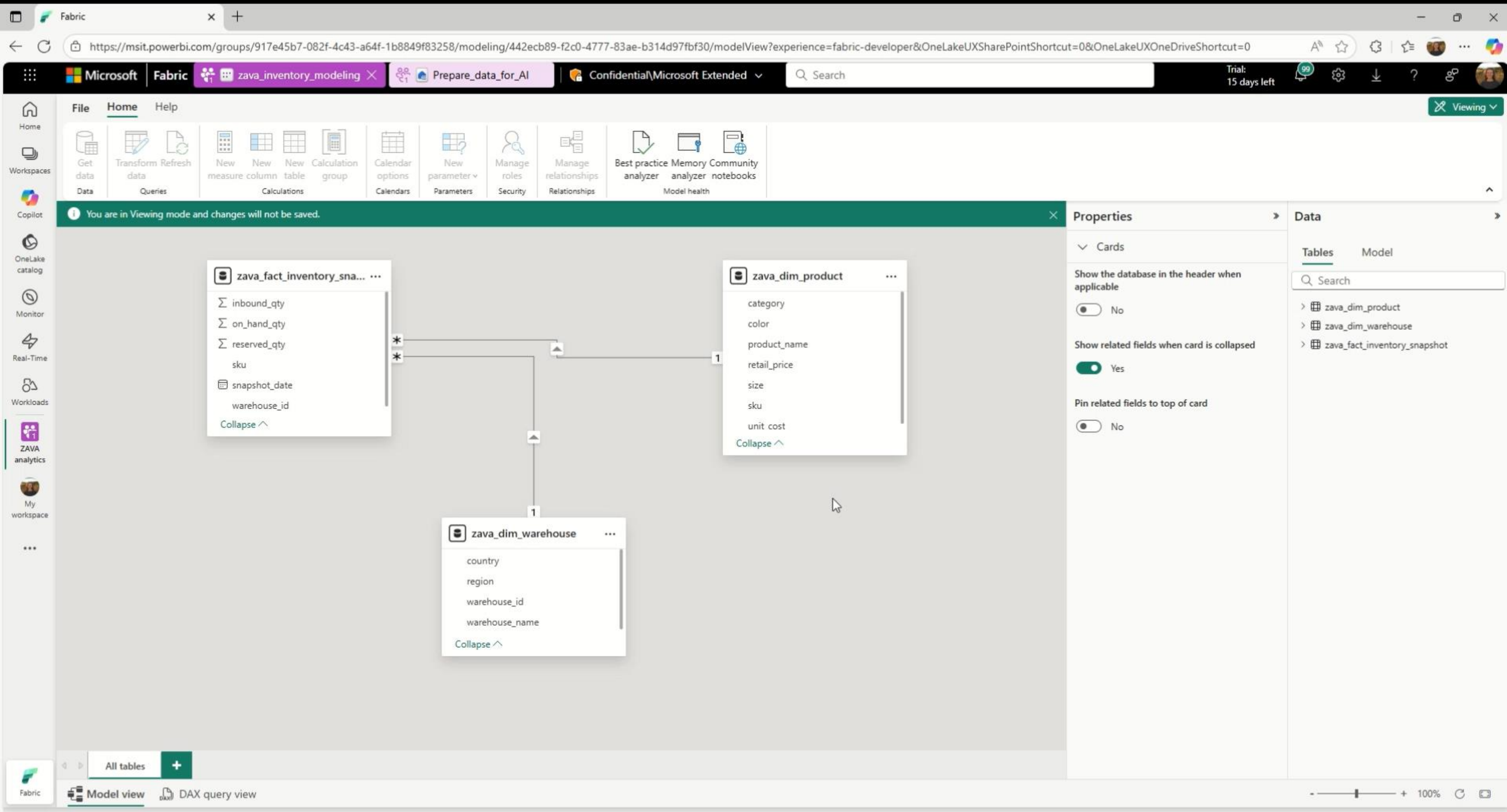


AI transformations built directly into OneLake

Extend easily by pointing to source folders and adding defined transformations

Right-click operations enabled on all folders across all clouds

Auto track changes in source and sync the destination



Fabric
https://msit.powerbi.com/groups/917e45b7-082f-4c43-a64f-1b8849f83258/modeling/442ecb89-f2c0-4777-83ae-b314d97fbf30/modelView?experience=fabric-developer&OneLakeUXSharePointShortcut=0&OneLakeUXOneDriveShortcut=0
Microsoft
Fabric
zava_inventory_modeling
Prepare_data_for_AI
Confidential\Microsoft Extended
Search
Trial: 15 days left
File
Home
Help
Viewing
Get data
Transform data
Refresh
New measure
New column
New table
Calculation group
Calendar options
New parameter
Manage roles
Manage relationships
Best practice analyzer
Memory analyzer
Community notebooks
Data
Queries
Calculations
Calendars
Parameters
Security
Relationships
Model health
You are in Viewing mode and changes will not be saved.
Home
Workspaces
Copilot
OneLake catalog
Monitor
Real-Time
Workloads
ZAVA analytics
My workspace
zava_fact_inventory_sna...
inbound_qty
on_hand_qty
reserved_qty
sku
snapshot_date
warehouse_id
Collapse
zava_dim_product
category
color
product_name
retail_price
size
sku
unit cost
Collapse
zava_dim_warehouse
country
region
warehouse_id
warehouse_name
Collapse
Properties
Cards
Show the database in the header when applicable
No
Show related fields when card is collapsed
Yes
Pin related fields to top of card
No
Data
Tables
Model
Search
zava_dim_product
zava_dim_warehouse
zava_fact_inventory_snapshot
All tables
Fabric
Model view
DAX query view
100%

Shortcut transformations

Seamless enrichment with AI in OneLake

Convert Formats

New

Parquet to Delta/Iceberg

New

JSON to Delta/Iceberg

CSV to Delta/Iceberg

Enhance with AI

Conversation

PII detection

Translate

Text summary

Sentiment

Name recognition

Key phrase extraction

Let’s talk catalogs!

Every system has them; databases, meta stores, services. They are a building block of a solution

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And now with dozens of catalogs, the focus is on making them work together

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OneLake is not only a storage or catalog service, but unifies these capabilities into an end-to-end unified data lake as a service

OneLake unified data foundation

Mirroring

Connect external catalogs (Operational DBs, Lakes, Business platforms)

Metadata is automatically synchronized (Tables, columns)

Data connected through shortcuts when stored in open storage or CDC replication and landed in open storage formats (storage is free)

Microsoft OneLake

OneLake security

Fabric

https://msit.powerbi.com/groups/917e45b7-082f-4c43-a64f-1b8849f83258/list?experience=fabric-developer

Microsoft | Fabric | Prepare_data_for_AI | Inventory Guardian | Search | Trial: 15 days left

Home · Workspaces · Copilot · OneLake catalog · Monitor · Real-Time · Workloads · ZAVA analytics · My workspace · Fabric

ZAVA analytics

Create deployment pipeline · Create app · Manage access · Workspace settings

+ New item · New folder · Import · Migrate · Filter by keyword · Filter

Name	Type	Task	Owner	Refreshed	Next refresh	Endorsement	Sensitivity	Included in app
Inventory Guardian	Data agent	—	Miquella de Boer	—	—	—	Confidential\Micro...	
Prepare_data_for_AI	Lakehouse	—	Miquella de Boer	—	—	—	Confidential\Micro...	
Prepare_data_for_AI	SQL analytics endp...	—	Miquella de Boer	—	—	—	Confidential\Micro...	
Transactions	Copy job	—	Miquella de Boer	—	—	—	Confidential\Micro...	
Zava modeling	Report	Store	ZAVA analytics	9/11/2025, 7:49:26 AM	—	—	Confidential\Micro...	No
zava_inventory_modeling	Semantic model	—	ZAVA analytics	9/11/2025, 7:49:26 AM	N/A	—	Confidential\Micro...	

Unify data in OneLake with zero ETL

Generally available

- Azure SQL DB
- Azure Data Lake Store
- Microsoft OneLake
- Google Cloud Storage
- Microsoft Dataverse
- Snowflake
- Amazon S3
- S3 Compatible (cloud/on-prem)
- Databricks Catalog
- Azure SQL MI (New)
- Azure Blob Storage (New)

Public preview

- Azure Cosmos DB
- Azure PostgreSQL
- Oracle DB (New)
- SQL Server
- Google BigQuery (New)
- SQL Server 2025

Microsoft OneLake

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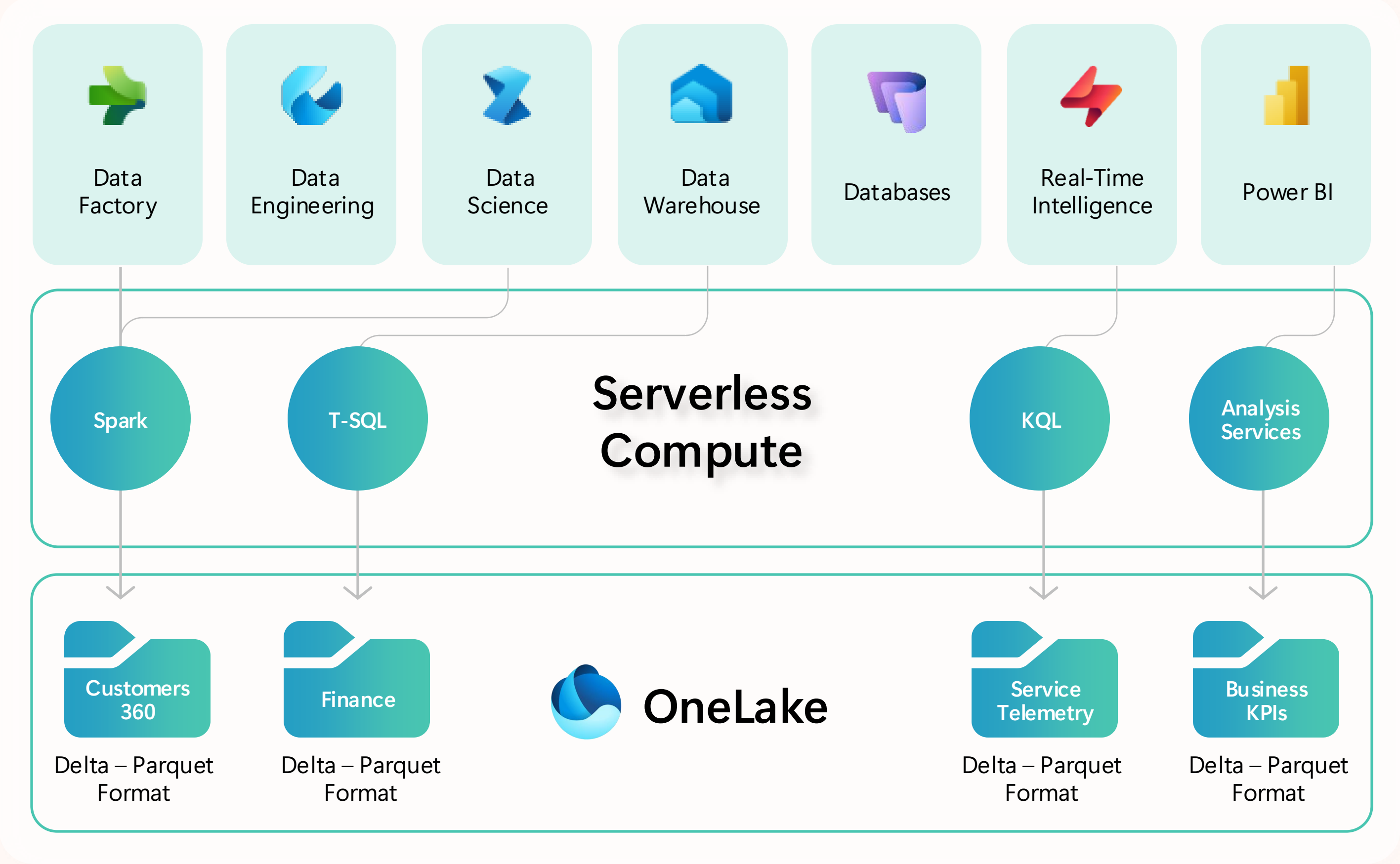
Unified discovery and governance

One copy for all computes



All the compute engines store their data automatically in OneLake as data items.

The data is stored in a single common format.

Delta – Parquet, an open standards format, and it is the storage format for all tabular data in Fabric.

All the compute engines have been fully optimized to work with Delta Parquet as their native format.

OneLake

Delta Lake Format

Iceberg Format

Transparent simultaneous support for Delta Lake and Iceberg formats

OneLake unified data foundation

Mirroring/ Shortcuts

Storage APIs

OneLake.DFS & .BLOB

Microsoft OneLake

OneLake Security

Public Preview:
OneLake.Table

Iceberg and delta
standard REST endpoints

Storage APIs
OneLake.DFS & BLOB

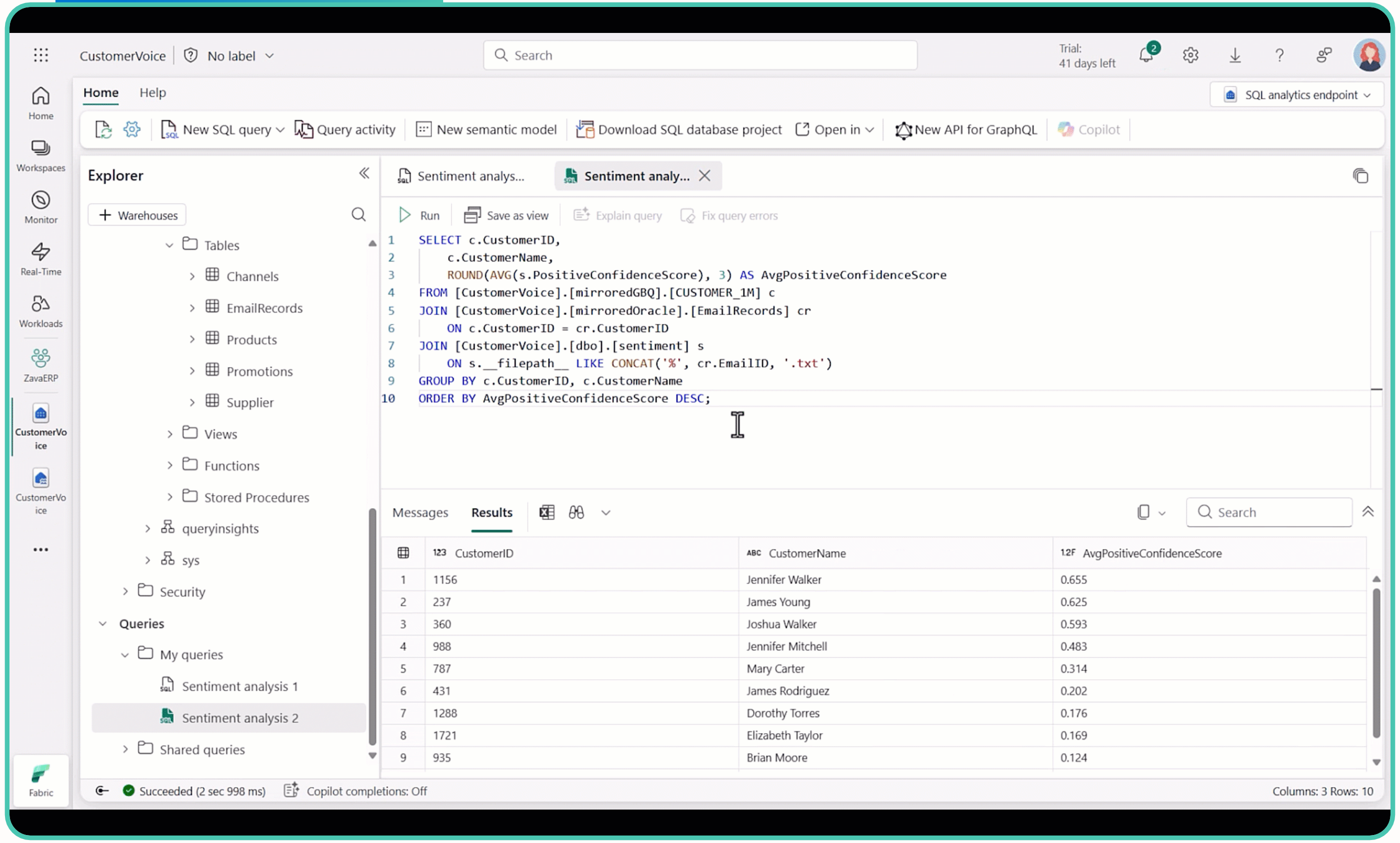
High scale APIs compatible with all Storage SDKs and applications

View data as blob or files

All data is available on all interfaces

OneLake.Table Endpoint

Coming Soon



Open-source table APIs directly atop your OneLake estate

Automatically available with no extra catalogs required

Compatible with **both Iceberg- and Delta**-compliant clients

Access **all OneLake tables** through **any interface**

Snowflake supports OneLake

aka.ms/Fabric-Snowflake

Available now

Snowflake support for native storage and bi-directional data access to OneLake

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Snowflake data accessible to all Fabric engines, Microsoft 365, Copilot

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Fabric Data accessible by Snowflake through catalog-linked databases

Microsoft OneLake Interoperability Partners

Data platforms

snowflake

dremio

Starburst

ClickHouse

Fivetran

Business platforms

EON

salesforce

servicenow

Open source

DuckDB

APACHE Spark

Polars

trino

pandas

Prepare_data_for_AI - Fabric | Link table API ZAVA - Snowflake

https://app.snowflake.com/east-us-2.azure/lca53710/w2cqvjsKRjgq#query

query | Link table API ZAVA

Databases | Worksheets

ACCOUNTADMIN · COMPUTE_WH (X-Small) | Share

Search objects

- MSIT_VALIDATIONTRACKERNEW2
- MSIT_VALIDATIONTRACKERNEW3
- MSIT_VALIDATIONTRACKERNEW4
- MSIT_VALIDATIONTRACKERSK
- MSIT_VALIDATIONTRACKER_KARTHIK
- NORTHEUROPE_VALIDATIONTRACKER
- ONELAKETEST
- SFDBDAILY0830
- SNOWFLAKE
- SNOWFLAKETESTDB20250911T0149
- SNOWFLAKE_LEARNING_DB
- SNOWFLAKE_SAMPLE_DATA
- TESTSNOWFLAKEDB
- ZAVA_INVENTORY
- ZAVA_IRC_CATALOG_LINKED
- ZAVA_PREP_FOR_AI_IRC_LINKED
  - DBO
    - Tables
      - CRM_MIRRORED
      - ERP_MIRRORED
      - FINANCE_SEMANTIC_MOD...
      - SALESTRANSACTIONS
      - WEBSITE_ANALYTICS_EVE...
      - ZAVA_DIM_PRODUCT
      - ZAVA_DIM_WAREHOUSE
      - ZAVA_FACT_INVENTORY_S...

No Database selected | Settings | Open in Workspaces | Code Versions

```sql
-- https://docs.snowflake.com/en/user-guide/tables-iceberg-configure-catalog-integration-rest
CREATE OR REPLACE CATALOG INTEGRATION ZAVA_PREP_FOR_AI_IRC_LINKED
  CATALOG_SOURCE = ICEBERG_REST
  TABLE_FORMAT = ICEBERG
  REST_CONFIG = (
    CATALOG_URI   = 'https://onelake.table.fabric.microsoft.com/iceberg'        -- onelake table endpoint for iceberg
    CATALOG_NAME  = 'ZavaERP/CustomerVoice.Lakehouse'                            -- Fabric data item, 'workspaceName/dataItemName'
  )
  REST_AUTHENTICATION = (
    TYPE                 = OAUTH                                                 -- Entra auth
    OAUTH_TOKEN_URI      = 'https://login.microsoftonline.com/81ba0d80-2361-40bb-9c1f-bf1c84027025/oauth2/v2.0/token'  -- tenant id
    OAUTH_CLIENT_ID      = '23c09802-d938-4745-bbd2-0e102fe62caa'                -- app client id
    OAUTH_CLIENT_SECRET  = 'HbeBQ-P3LaXIM-s$xsevZ1qZd0fhXxsRiXqQgdan'            -- secret value
    OAUTH_ALLOWED_SCOPES = ('https://storage.azure.com/.default')                -- storage token
  )
  ENABLED = TRUE
;
```

Results | Chart

	status
1	Integration ZAVA_PREP_FOR_AI_IRC_LINKED successfully created.

Query Details

Query duration 207ms

Rows 1

Query ID 01bf0249-0002-aa48-0...

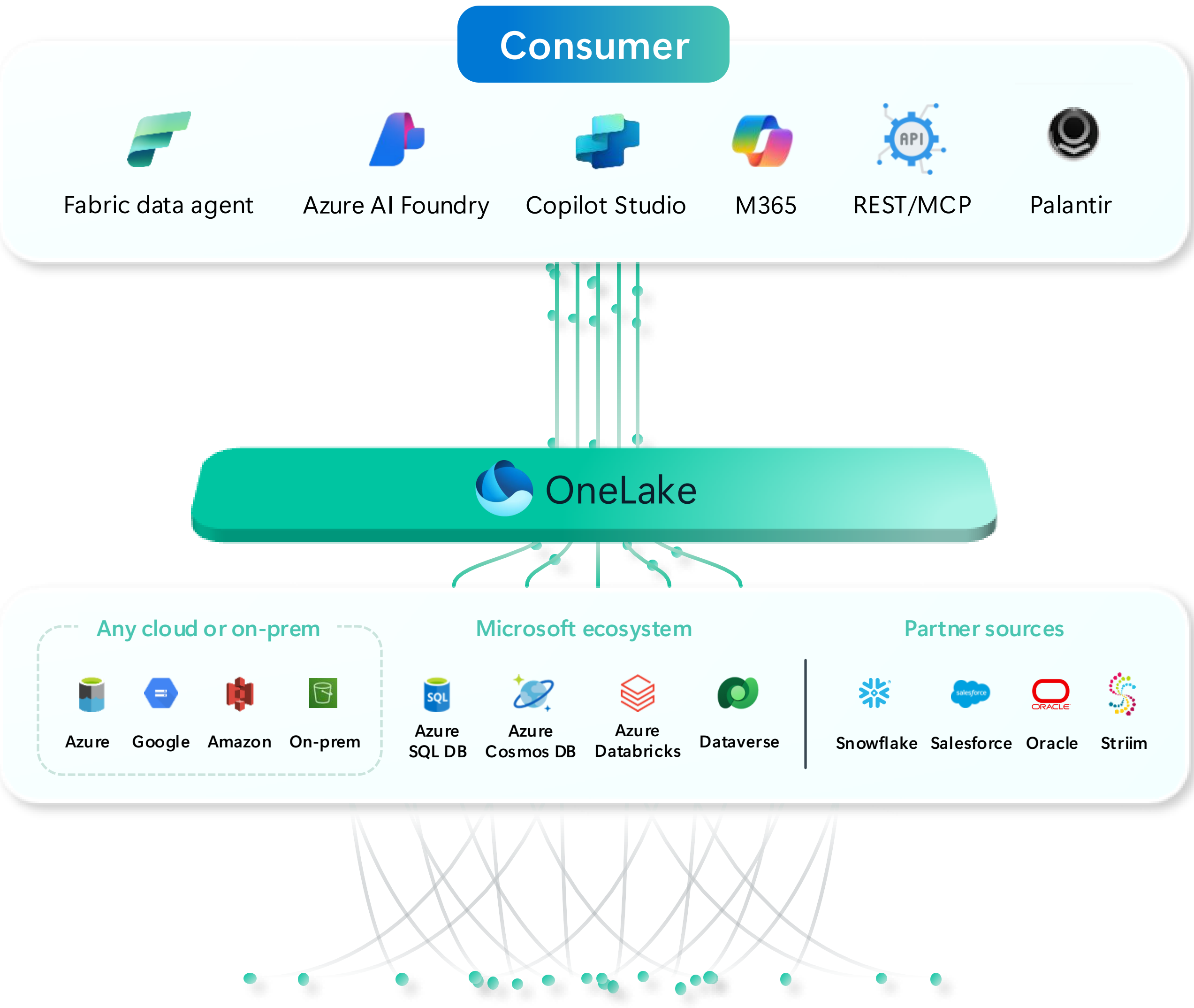
Show more

status

100% filled

Ask Copilot

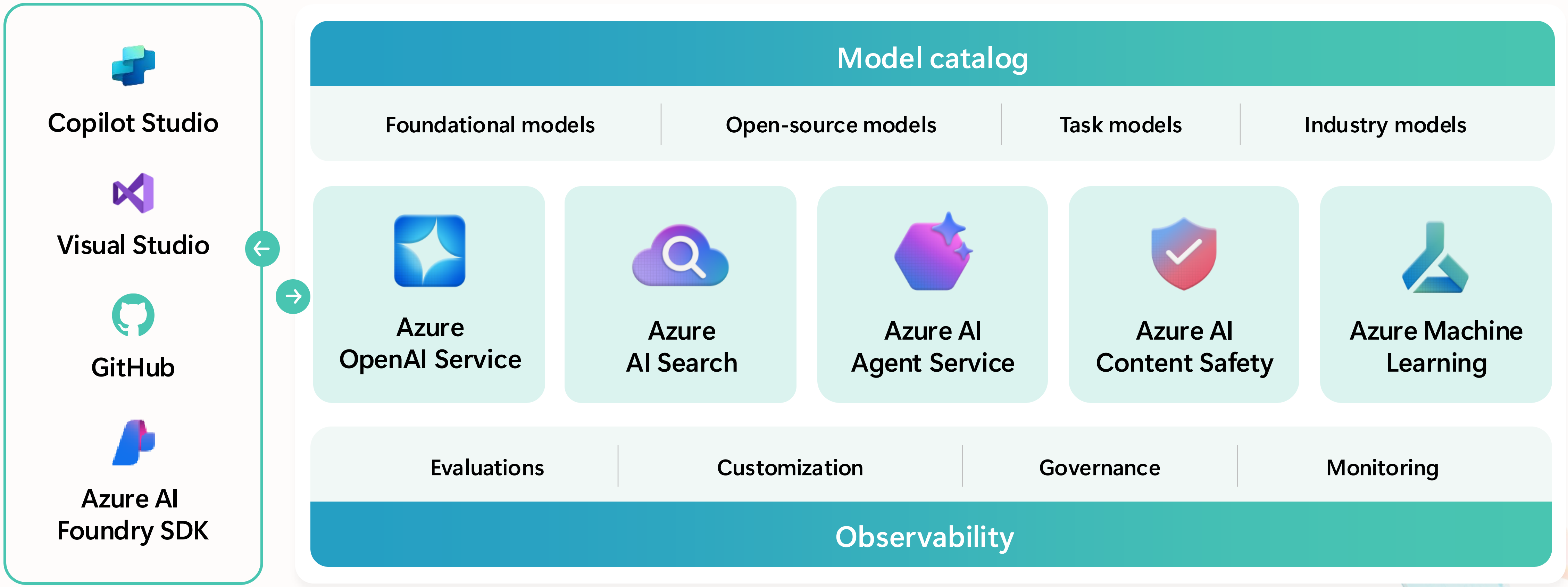
Unified data lake for AI



Unify and prep data for AI in OneLake

Connect to any **AI tool** in Fabric, Microsoft or Partners

Azure AI Foundry



INVOICE

No: INV-101-0828
Date: 2025-08-28
Supplier VAT: ATU99887766

Bill To
Zava GmbH, Wiedner Gürtel 11, 1040 Vienna, AT

Ship To
Warehouse 1 (Vienna DC), Triester Str. 120, 1100 Vienna, AT

Supplier
AlpenTextil GmbH

Item		Description	Qty	Unit Cost	Line Total
	DIR-ALM-RED-38	Almsee Dirndl (Red)	60	88.00	5280.00
	DIR-ALM-BLU-40	Almsee Dirndl (Blue)	40	88.00	3520.00
				Subtotal	8800.00
				VAT 20%	1760.00
				TOTAL	**10560.00**

Thank you for your business. Payment due in 30 days.

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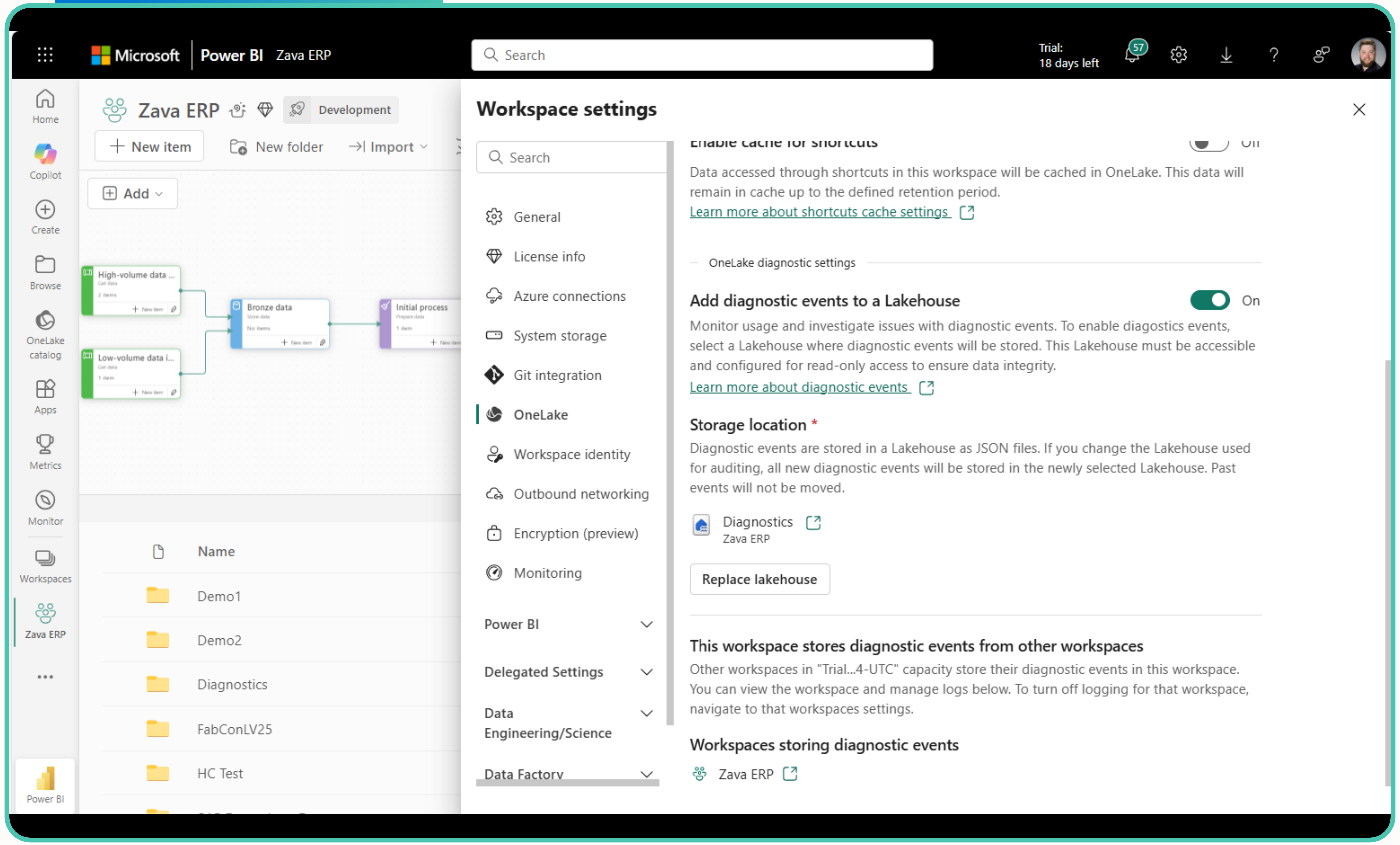
OneLake diagnostics

Coming Soon



Compatible with the Azure **storage diagnostic** feature

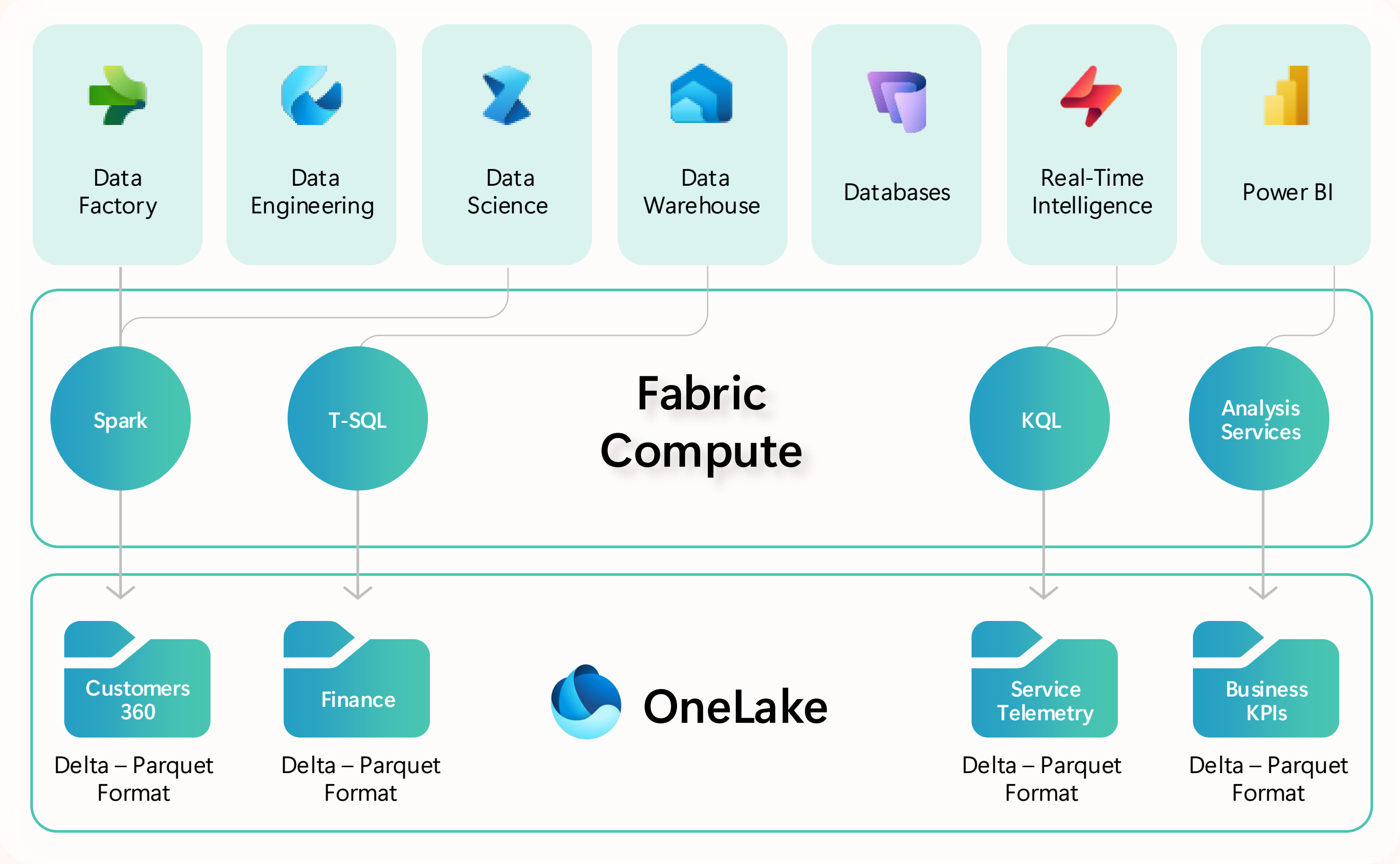
Spans **all data** stored in OneLake, mirrored, or virtualized through shortcuts

Tracks performance, latency, **availability,** capacity, **errors,** and **request** metrics.

Gives a detailed picture of **storage usage** across OneLake

OneLake security

Defined once, enforced everywhere



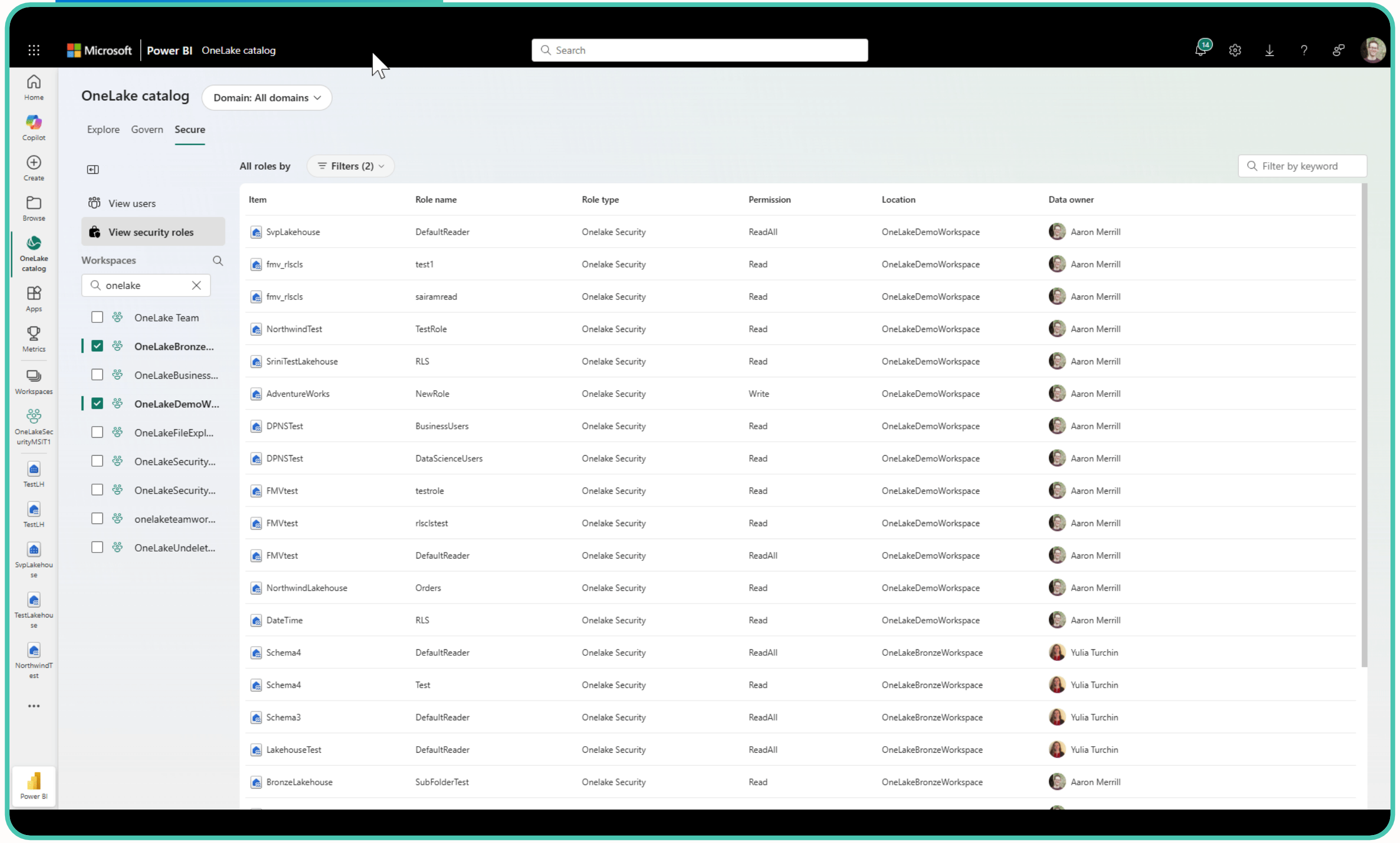
Define security roles in OneLake using powerful features like table, row, or column level security.

Data is secured consistently across experiences inside and outside of Fabric.

OneLake security roles can be managed in a single place, providing end to end coverage for data access for your entire data estate.

OneLake security updates

Coming Soon



OneLake security is soon **available to all customers** in public preview

Centrally view and manage security in the new OneLake catalog **secure tab**

Improved **REST APIs** for automation

All mirrored items are now supported

Gain full visibility and control with industry-leading features

Governance and security features built into Microsoft Fabric

Manage your data estate

Admin portal
Tenant and workspace settings
Metadata scanning
Capacities
Domains
Workspaces
Lifecycle management

Secure, protect, and comply

Fabric inbound and outbound security
Conditional access
Compute security
Workspace & item security
Data encryption
OneLake security
Purview labeling**
Purview DLP policies**
Certificates & standards
Data residency
Purview Audit**
Purview DSPM for AI **

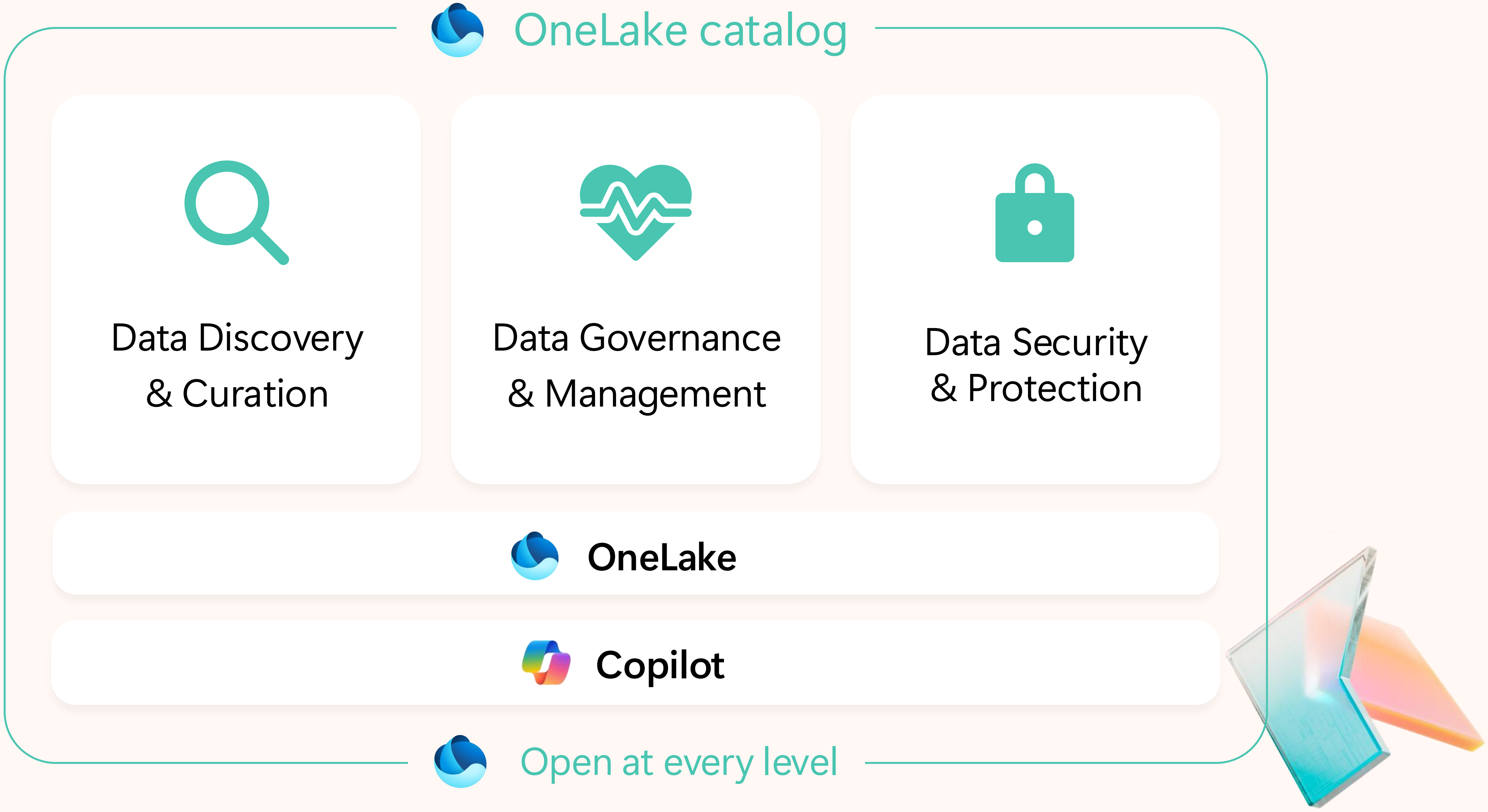
Encourage data discovery, trust, and use

Discovery w/ OneLake catalog
Endorsement & tags
Data lineage & impact analysis
External data sharing
Metadata curation
Discover & Reuse
DQ via 1p/3p**

Monitor, uncover insights, and act

Govern w/ OneLake catalog
Monitoring hub
Workspace monitoring
Capacity metrics
Admin monitoring
OneLake Diagnostics

One catalog to centrally Discover, Govern, Protect



One catalog to centrally Discover, Govern, Protect

OneLake catalog

Data Discovery & Curation

Data Governance & Management

Data Security & Protection

OneLake

Copilot

Open at every level

Leverage open standards with Delta/Parquet format & data access

Access programmatically with Admin & User APIs

1st & 3rd party catalog integrations

Any data, anywhere

Discover & Explore all Fabric item types, 1st and 3rd party

Built into office applications and overall 100+ scenarios in Fabric and beyond

Data prep for AI & AI-Powered data curation & insights

Built-in governance at Scale

OOTB governance insights & proactive recommendations

Implement a data mesh by grouping into business domains

[Coming soon] Fabric policies to enforce Org protection & compliance; Central management for Admin roles ;

Zero-trust

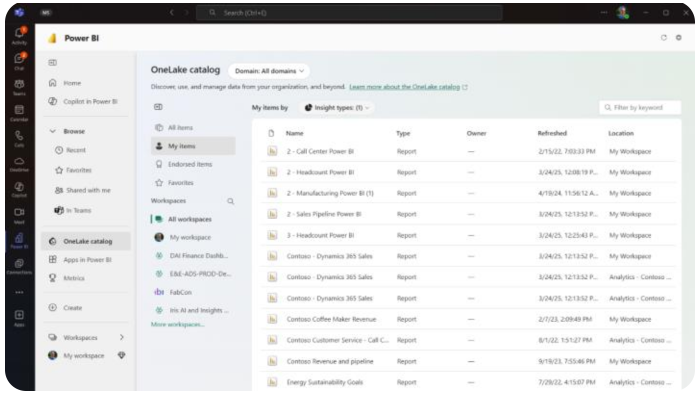
Catalog as the central place for OneLake Security Management

Data protection with Purview security (MIP, DLP)

Secure AI usage in Fabric, and ensure data is AI ready

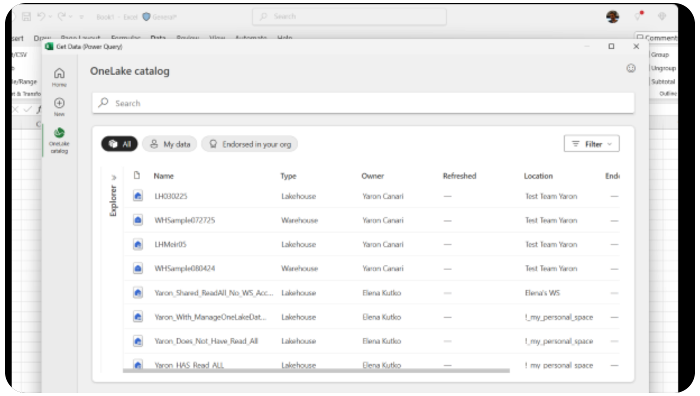
OneLake catalog is built into Microsoft 365

Bringing data access to the 350 million Microsoft 365 users



Microsoft Teams

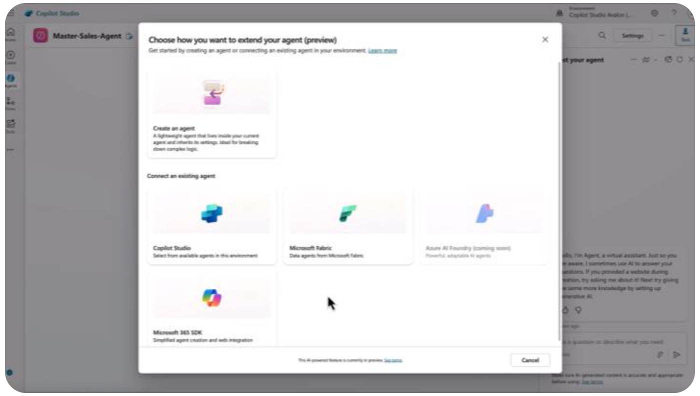
Combine OneLake with Teams to infuse data into everyday work with embedded channel, chat, and meeting experiences



Microsoft Excel

Access the entire OneLake catalog in Excel and make live changes in your workbooks without copying or moving the data

*Insider Slow ring



Microsoft Copilot Studio

AI creators can enrich custom copilot agents in Copilot Studio with data from OneLake

Plus 100+ other scenarios

Azure AI Foundry

SharePoint

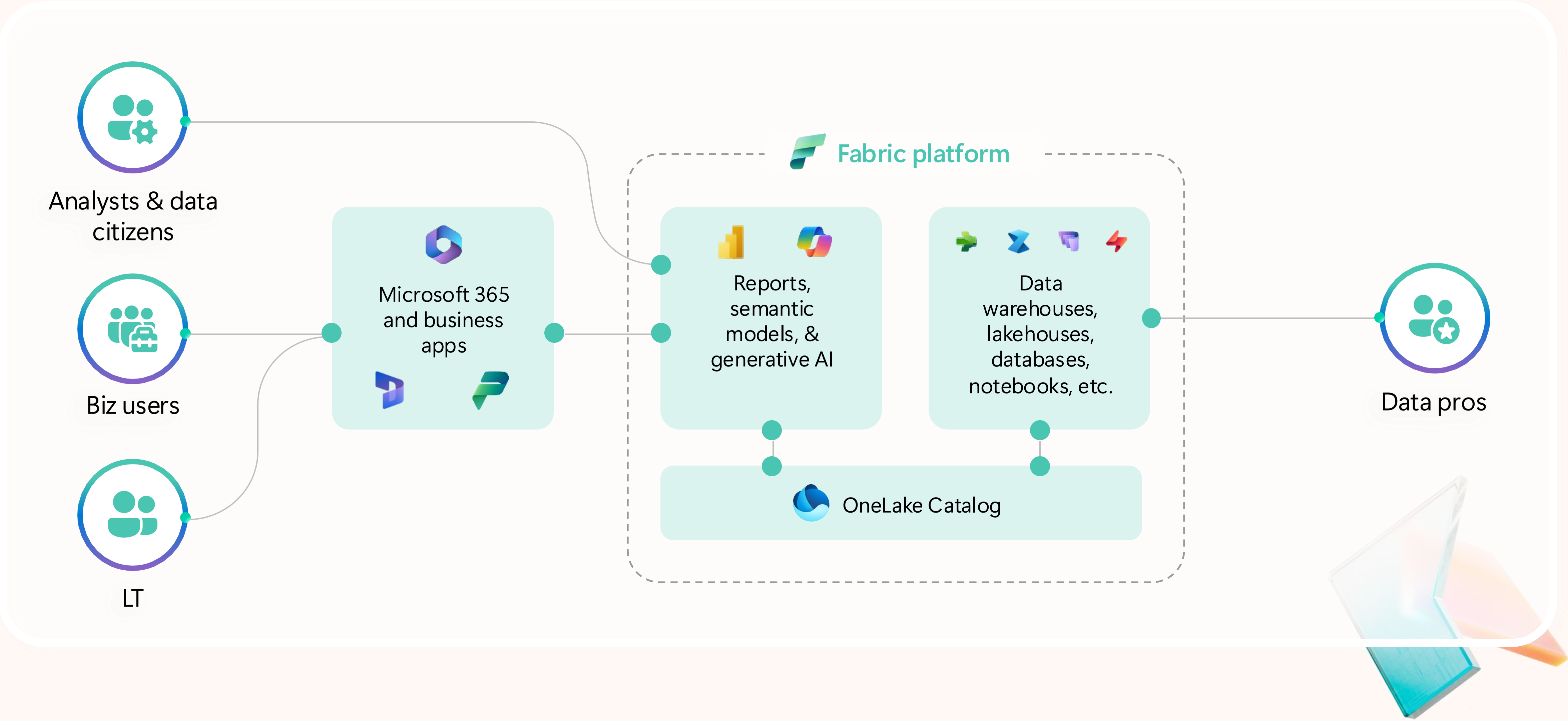
OneDrive

Outlook

Dynamics 365

Power Platform

AI-powered data cultures require broad data access



The trusted catalog for organizations worldwide

The default source of data for business users

30M+

monthly active Power BI and Fabric users can access their data through the OneLake catalog

Used by 95% of the Fortune 500

230K

organizations store, manage, and govern their data in the OneLake catalog

Accessible from anywhere

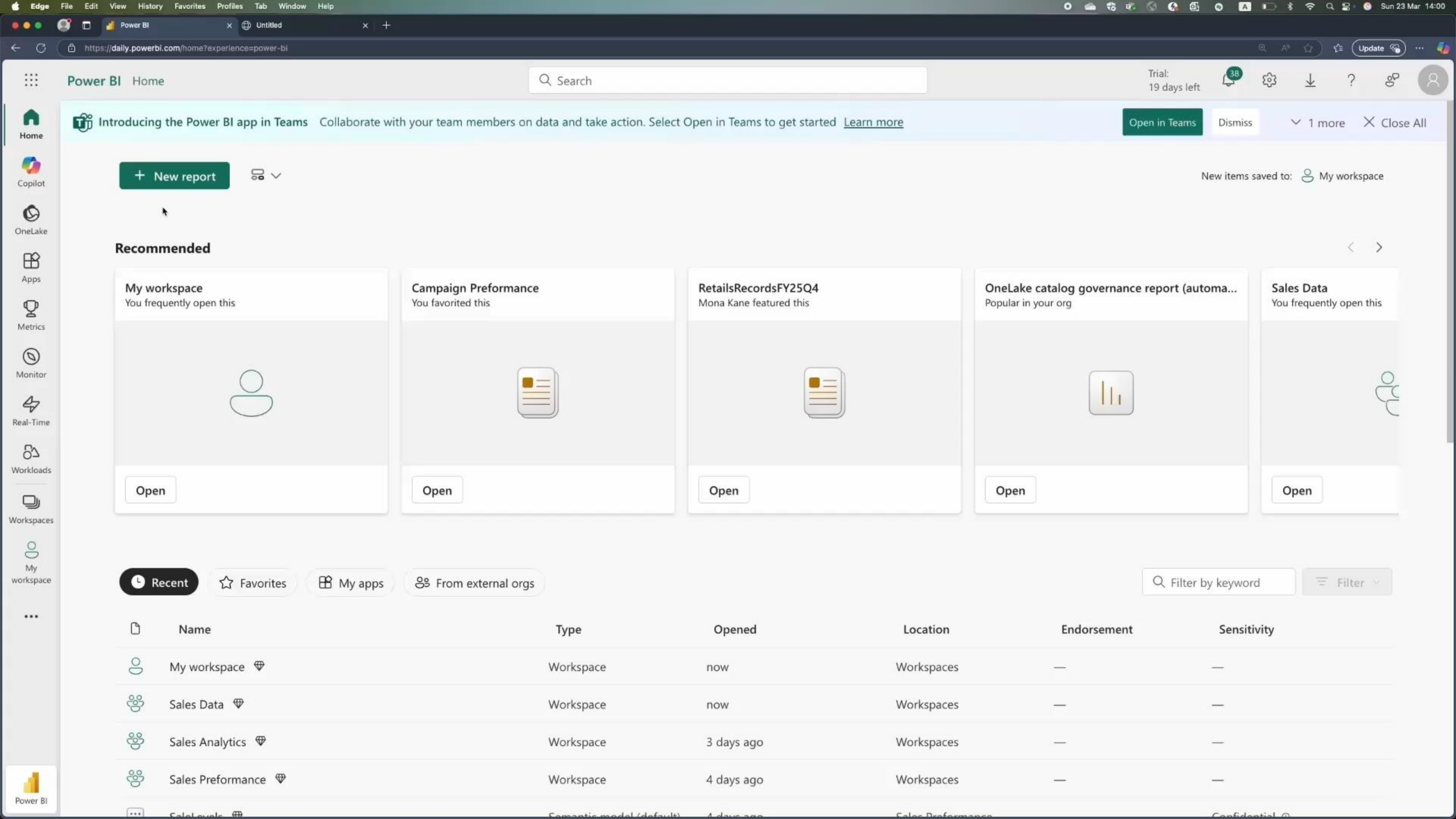
100+

of the most widely used apps have OneLake catalog embedded like Microsoft Teams and Excel

*Comparative analysis based on publicly disclosed customer count and third-party estimates

Demo

OneLake catalog and security


Edge File Edit View History Favorites Profiles Tab Window Help
Sun 23 Mar 14:00
Power BI
Untitled
https://daily.powerbi.com/home?experience=power-bi
Update
Power BI Home
Search
Trial:
19 days left
38
Home
Copilot
OneLake
Apps
Metrics
Monitor
Real-Time
Workloads
Workspaces
My workspace
Power BI
Introducing the Power BI app in Teams
Collaborate with your team members on data and take action. Select Open in Teams to get started
Learn more
Open in Teams
Dismiss
1 more
Close All
New report
New items saved to:
My workspace
Recommended
My workspace
You frequently open this
Open
Campaign Preformance
You favorited this
Open
RetailsRecordsFY25Q4
Mona Kane featured this
Open
OneLake catalog governance report (automa...
Popular in your org
Open
Sales Data
You frequently open this
Open
Recent
Favorites
My apps
From external orgs
Filter by keyword
Filter
Name
Type
Opened
Location
Endorsement
Sensitivity
My workspace
Workspace
now
Workspaces
—
—
Sales Data
Workspace
now
Workspaces
—
—
Sales Analytics
Workspace
3 days ago
Workspaces
—
—
Sales Preformance
Workspace
4 days ago
Workspaces
—
—

3rd Annual Fabric Community Conference in Europe

Barcelona, Spain

28th September – 1st October 2026

aka.ms/FabCon-EU